

VITYARTHIPROJECT

To-Do List Manager

Overview

To-Do List Manager is a command-line Python application that helps users add, organize, and track their daily tasks. Each task has a description, priority level, and due date, and can be marked complete, edited, or deleted. Tasks are automatically saved to a local file, so nothing is lost between runs of the program. The project was built for the Introduction to Programming and Solving course to apply core programming concepts — functions, conditionals, loops, dictionaries, and file handling — in a complete, usable application.

Features

- Add a new task with a description, priority (High/Medium/Low), and due date
- View all tasks in a numbered, readable list showing status
- Mark a task as complete
- Update a task's description, priority, or due date
- Delete a task
- Filter tasks by status (pending/completed only)

- Filter tasks by priority
- Sort tasks by due date
- View summary counts (total, completed, pending)
- All data is saved automatically to a local JSON file

Technologies / Tools Used

- Python 3
- Built-in json module (for saving/loading task data)
- Built-in os module (for checking if the data file exists)
- Git & GitHub (for version control and submission)

Steps to Install & Run

1. Make sure Python 3 is installed on your computer. Check by running:

`python --version`

2. Clone or download this repository.
3. Open a terminal in the project folder.
4. Run the program:

`python main.py`

5. Use the on-screen numbered menu to add, view, update, delete, or filter your tasks.

Instructions for Testing

1. Run python main.py.
2. Choose option 1 to add a few tasks with different priorities and due dates.
3. Choose option 2 to confirm the tasks appear correctly in the list.
4. Choose option 3 to mark a task complete or edit one, then view the list again to confirm the change was saved.
5. Choose option 4 to delete a task, then view the list again to confirm it's gone.
6. Choose option 5 to try each report/filter (pending only, completed only, by priority, sorted by due date, summary counts).
7. Close the program and re-run it — confirm your previously added tasks are still there (loaded from tasks.json).
8. Try an invalid input (e.g. a letter instead of a task number) to confirm the program shows an error message instead of crashing.

Screenshots

